

Chapter 4. Three Axis Active Control

Three axis active control system consists of two main classes

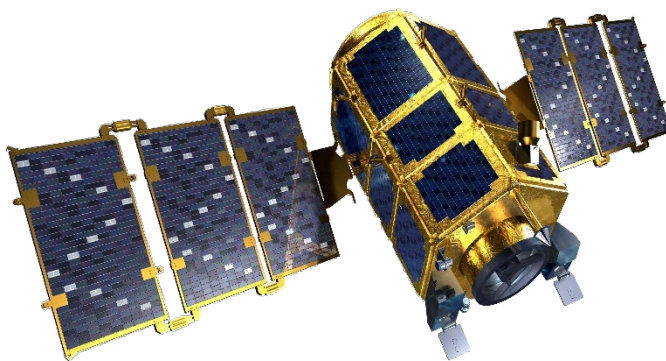
- 1) mass expulsion (pure jet)
- 2) angular momentum exchange (RW, CMG)

Advantages: Fast response,
Good pointing accuracy,
Possibility of using solar panels,
Noninertial pointing capability,
The addition of momentum exchange devices prolongs the life of the system

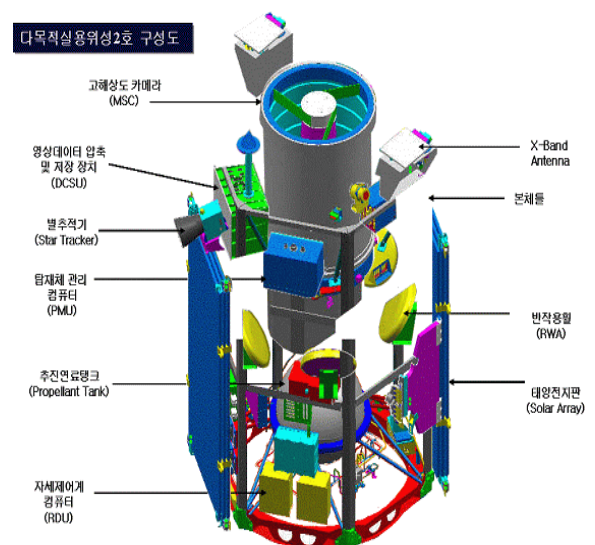
Disadvantages: limited due to fuel expenditure,
Scanning sensors,
No good performances due to gas jet failure,
Complexity due to momentum exchange devices

Typical sequence of control operations

- i) acquire sun
- ii) acquire earth and star
- iii) establish normal limit cycle
- iv) perform reorientation
- v) perform control



KOMPSAT2 (2006.7.28, 1m image)



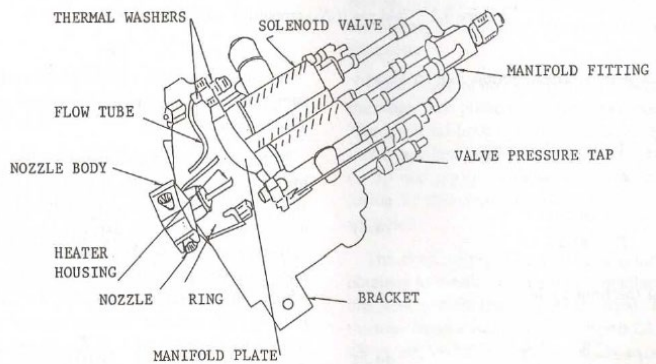
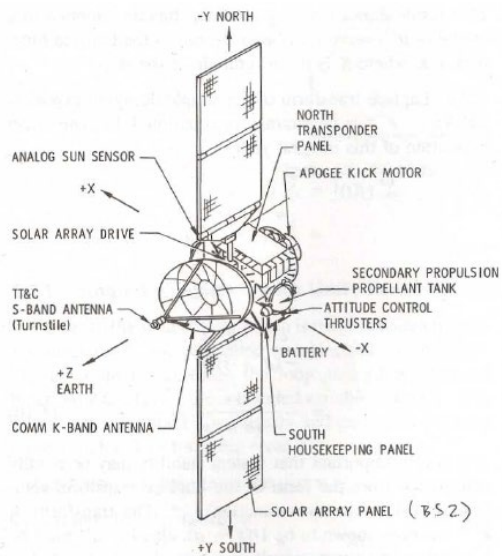


Figure 4.4 An example of a low-level attitude control thruster assembly.

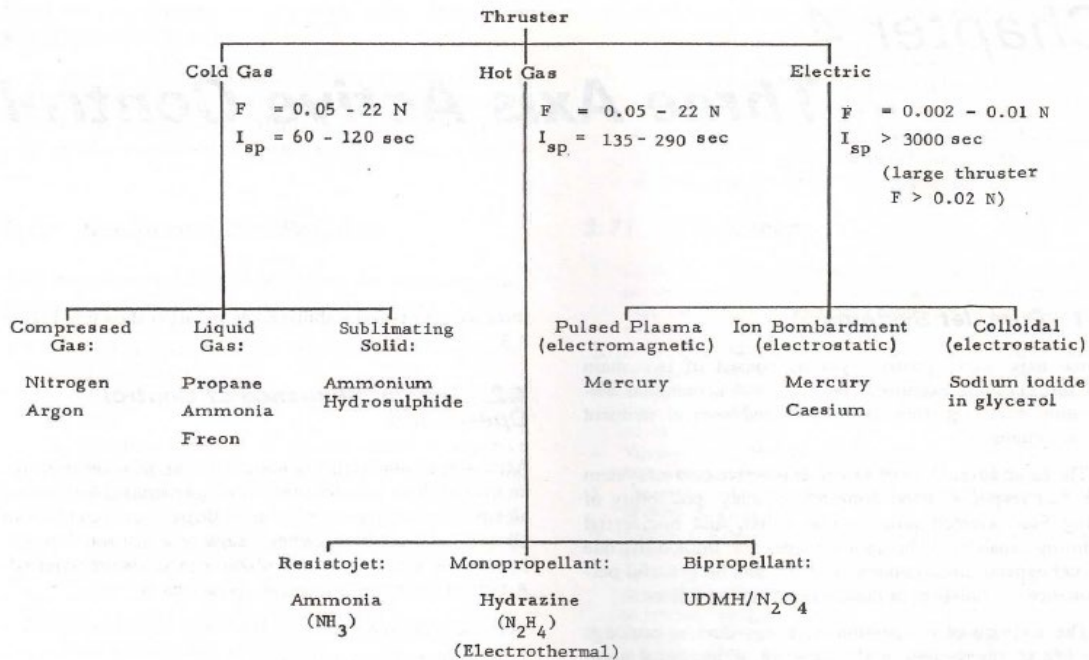


Figure 4.1 Mass expulsion thrusters.

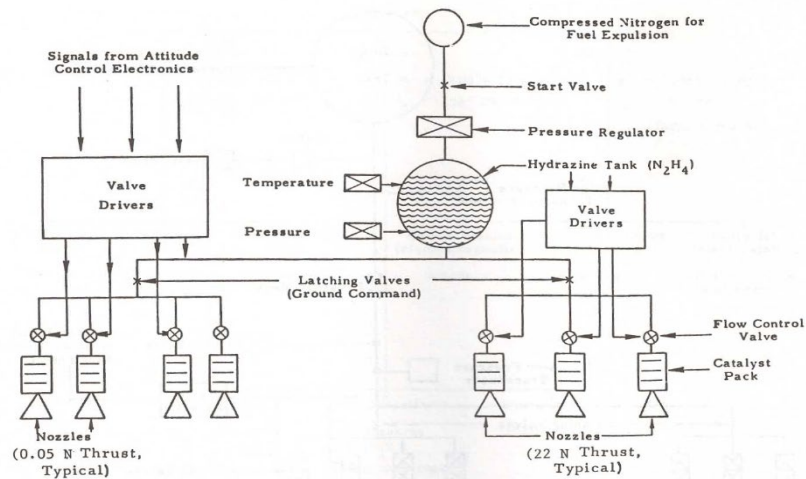


Figure 4.3 Hydrazine propulsion subsystem.

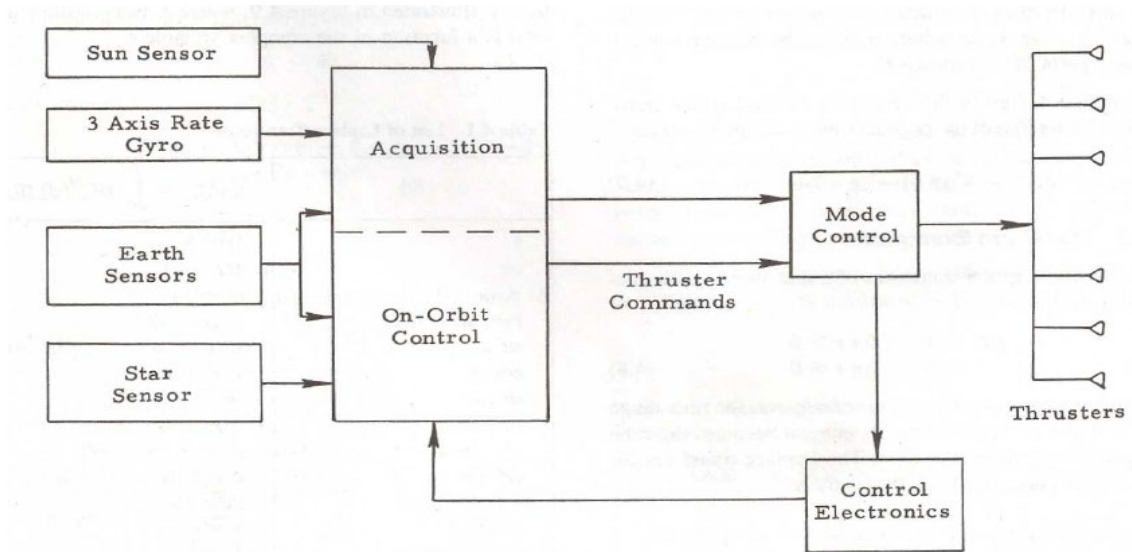
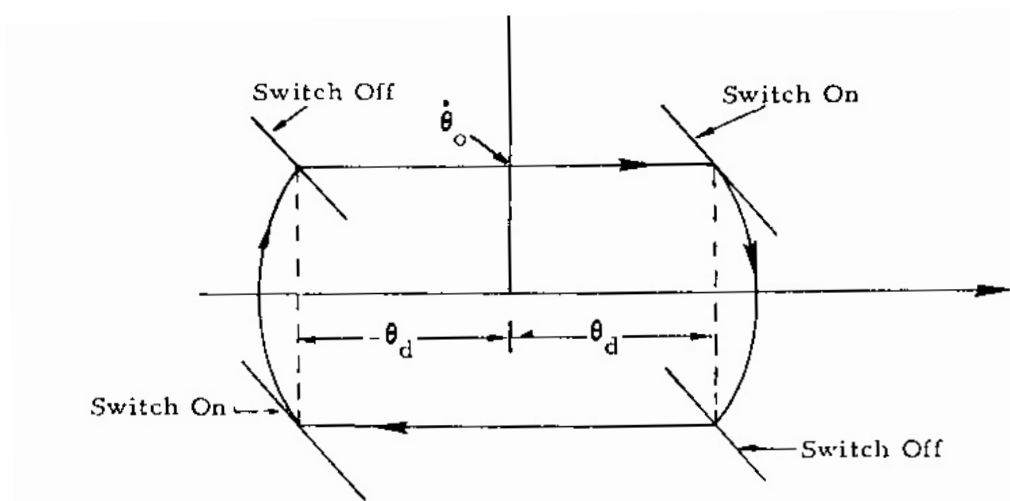


Figure 4.6 A typical functional attitude control system block diagram.

On orbit operations

The attitude angles about each of the three orthogonal axes are required to stay within prescribed limits(dead band) using thrusters.

The angular velocity within the dead band may also be specified but is generally required to be as small as possible to prevent interference with the spacecraft payload functions and to minimize fuel consumption.



A constant $\dot{\theta}_0$ exists between the negative ($-\theta_d$) and positive (θ_d) values of the dead band. Minimization of this velocity results in the improved gas consumption. The average propellant consumption rate can be obtained by finding the ratio of propellant expended during one limit cycle to the limit cycle period. The total angular momentum change is

$$\Delta h = 2I(\Delta\dot{\theta})$$

The propellant mass expended is

$$m = \frac{I_p}{gI_{sp}} = \frac{F\Delta t}{gI_{sp}}$$

$$\text{where } Fl = \frac{\Delta h}{\Delta t}$$

I = moment of inertia

I_{sp} = specific impulse

l = jet moment arm about the vehicle center of mass

$$= \frac{\Delta h}{gI_{sp}l}$$

$$\dot{m} = \frac{1}{P} \frac{\text{propellant mass}}{\text{cycle}}$$

$$= \frac{1}{P} \frac{2I(\Delta\dot{\theta})}{gI_{sp}l}$$

where P = period of the limit cycle

$$P = \frac{4\theta_d}{\dot{\theta}_0} = \frac{8\theta_d}{\Delta\dot{\theta}} \left(\Delta\dot{\theta} = 4\dot{\theta}_0 = \frac{\Delta h}{2I} = \frac{Fl\Delta t}{2I} \right)$$

$$\dot{m} = \frac{\Delta\dot{\theta}}{8\theta_d} \frac{2I\Delta\dot{\theta}}{gI_{sp}l}$$

$$= \frac{2I}{8\theta_d} \frac{(Fl\Delta t)^2}{4I^2} \frac{1}{gI_{sp}l}$$

$$= \frac{(Fl\Delta t)^2}{16IgI_{sp}l\theta_d}$$

The propellant consumption rate is minimized for a small torque impulse and a large deadband.

Servomechanisms

Servomechanisms(or servos) are closed-loop control system used to determine the position, velocity or acceleration of mechanical loads.

An open loop refers to system for which the output variable is not measured nor is this information fed back to affect the system operation.

In a closed loop system, the output variable is measured and compared with the input command; the difference between the two is used to make the system correct itself.

The basic properties of servos are revealed through the use of linear differential equations with constant coefficients.

Laplace transform

$$F(s) = L[f(t)] = \int_0^{\infty} f(t)e^{-st} dt$$

linear differential equation $\xrightarrow{\text{Laplace transform}}$ *algebraic equation*

$$\text{ex. } f(t) = e^{-at}$$

$$\begin{aligned} L[f(t)] &= L[e^{-at}] \\ &= \int_0^{\infty} e^{-at} e^{-st} dt \\ &= \int_0^{\infty} e^{-(a+s)t} dt \\ &= \left[-\frac{e^{-(a+s)t}}{(s+a)} \right]_0^{\infty} \\ &= \frac{1}{(s+a)} \end{aligned}$$

<Some Laplace Transform Theorems>

1. Definition

$$L[f(t)] = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

2. The linearity of the Laplace Transform:

$$L[af(t) + bg(t)] = aF(s) + bG(s)$$

3. The Laplace transform of the derivatives of a function:

$$L\left[\frac{df}{dt}\right] = sF(s) - f(0)$$

$$L\left[\frac{d^2 f}{dt^2}\right] = s^2 F(s) - sf(0) - \frac{df}{dt}(0)$$

$$L\left[\frac{d^n f}{dt^n}\right] = s^n F(s) - s^{n-1} f(0) - s^{n-2} \frac{df}{dt}(0) - s^{n-3} \frac{d^2 f}{dt^2}(0) - \dots - s \frac{d^{(n-2)} f}{dt^{(n-2)}}(0) - \frac{d^{(n-1)} f}{dt^{(n-1)}}(0)$$

4. The first shifting theorem:

$$L[e^{-at} f(t)] = F(s + a)$$

this is also commonly used in the form:

$$L^{-1}[F(s + a)] = e^{-at} L^{-1}[F(s)]$$

5. The second shifting (or translation) theorem:

$$L[f(t - T)] = e^{-sT} F(s)$$

6. The derivative of the Laplace transform of a function:

$$L[tf(t)] = -\frac{dF}{ds}$$

7. The initial value theorem:

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} [sF(s)]$$

8. The final value theorem:

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} [sF(s)]$$

9. The convolution theorem:

$$\int_0^t f(t-u)g(u)du = \int_0^t f(u)g(t-u)du = F(s)G(s)$$

<Table of Laplace transform pairs>

$F(s)$	$f(t)$
1	Unit impulse $\delta(t)$
$\frac{1}{s}$	Unit step 1(t)
$\frac{1}{s^2}$	t
$\frac{1}{s^n}$	$\frac{t^n}{(n-1)!}$ (n = 1, 2, 3...)
$\frac{n!}{s^{n+1}}$	t^n (n = 1, 2, 3...)
$\frac{1}{\sqrt{s}}$	$\frac{1}{\sqrt{\pi t}}$
$\frac{1}{s^{3/2}}$	$\frac{2}{\sqrt{t/\pi}}$
$\frac{1}{s^a}$	$\frac{t^{a-1}}{\Gamma(a)}$
$\frac{1}{(s+a)^k} (k > 0)$	$\frac{1}{\Gamma(k)} t^{k-1} e^{-at}$
$\frac{1}{s+a}$	e^{-at}
$\frac{1}{(s+a)^2}$	te^{-at}
$\frac{1}{(s+a)^n}$	$\frac{1}{(n-1)!} t^{n-1} e^{-at}$ (n = 1, 2, 3...)
$\frac{n!}{(s+a)^{n+1}}$	$t^n e^{-at}$ (n = 1, 2, 3...)
$\frac{1}{s^2 + \omega^2}$	$\frac{1}{\omega} \sin \omega t$
$\frac{s}{s^2 + \omega^2}$	$\cos \omega t$
$\frac{\omega}{s^2 + \omega^2}$	$\sin \omega t$
$\frac{s}{s^2 - \omega^2}$	$\cosh \omega t$
$\frac{\omega}{s^2 - \omega^2}$	$\sinh \omega t$
$\frac{1}{s(s+a)}$	$\frac{1}{a} (1 - e^{-at})$

$\frac{1}{(s+a)(s+b)}$	$\frac{1}{b-a}(e^{-at} - e^{-bt})$
$\frac{s}{(s+a)(s+b)}$	$\frac{1}{b-a}(be^{-bt} - ae^{-at})$
$\frac{1}{s(s+a)(s+b)}$	$\frac{1}{ab}\left[1 + \frac{1}{a-b}(be^{-bt} - ae^{-at})\right]$
$\frac{1}{s(s+a)^2}$	$\frac{1}{a^2}(1 - e^{-at} - ate^{-at})$
$\frac{1}{s^2(s+a)}$	$\frac{1}{a^2}(at - 1 + e^{-at})$
$\frac{\omega}{(s+a)^2 + \omega^2}$	$e^{-at} \sin \omega t$
$\frac{s+a}{(s+a)^2 + \omega^2}$	$e^{-at} \cos \omega t$
$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$	$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_n \sqrt{1-\zeta^2} t$
$\frac{s}{s^2 + 2\zeta\omega_n s + \omega_n^2}$	$-\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t - \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$
$\frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$	$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \phi)$ $\phi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$
$\frac{1}{s(s^2 + \omega^2)}$	$\frac{1}{\omega^2}(1 - \cos \omega t)$
$\frac{1}{s^2(s^2 + \omega^2)}$	$\frac{1}{\omega^3}(\omega t - \cos \omega t)$
$\frac{\omega^2}{s(s^2 + \omega^2)}$	$1 - \cos \omega t$
$\frac{\omega^3}{s^2(s^2 + \omega^2)}$	$\omega t - \sin \omega t$
$\frac{2\omega^3}{(s^2 + \omega^2)^2}$	$\sin \omega t - \omega t \cos \omega t$
$\frac{s}{(s^2 + \omega^2)^2}$	$\frac{1}{2\omega} t \sin \omega t$
$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$	$t \cos \omega t$

$\frac{s}{(s^2 + \omega_1^2)(s^2 + \omega_2^2)}$	$\frac{1}{\omega_2^2 - \omega_1^2}(\cos \omega_1 t - \cos \omega_2 t), \omega_1^2 \neq \omega_2^2$
$\frac{1}{(s^2 + \omega^2)^2}$	$\frac{1}{2\omega^3}(\sin \omega t - \omega t \cos \omega t)$
$\frac{s^2}{(s^2 + \omega^2)^2}$	$\frac{1}{2\omega}(\sin \omega t - \omega t \cos \omega t)$
$\ln \frac{s+a}{s+b}$	$\frac{1}{t}(e^{-bt} - e^{-at})$
$\ln \frac{s^2 + \omega^2}{s^2}$	$\frac{2}{t}(1 - \cos \omega t)$
$\ln \frac{s^2 - a^2}{s^2}$	$\frac{2}{t}(1 - \cosh at)$

Transfer Function

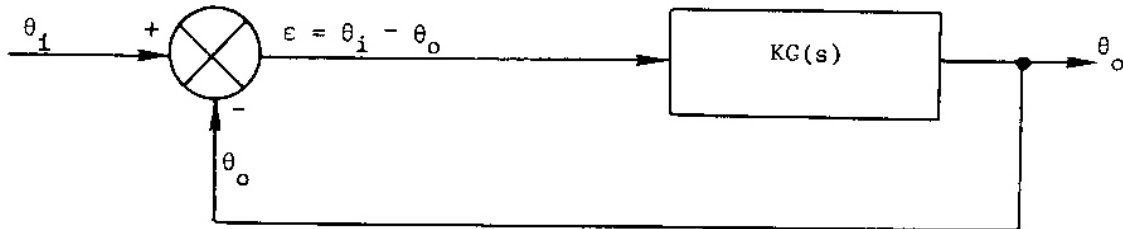
In control theory, functions called 'transfer functions' are commonly used to characterize the input-output relationships of components or systems that can be described by linear, time-invariant, differential equations.

The transfer function is defined the ratio of the Laplace transform of the output to the Laplace transform of the input under the assumption that all initial conditions are zero.

1. The transfer function of a system is a mathematical model in that it is an operational method of expressing the differential equation that relates the output variables to the input variable.
2. The transfer function is a property of a system itself, independent of the magnitude and nature of the input or driving function.
3. It does not provide any information concerning the physical structure of the system. So, the transfer functions of many physically different systems can be identical.
4. If the transfer function of a system is known, the output or response can be studied for various forms of inputs with a view toward understanding the nature of the system.
5. A transfer function gives a full description of the dynamic characteristics of the system, as distinct from its physical description.

A block diagram of a system is a pictorial representation of the functions performed by each component and of the flow of signals.

Single Loop Servo with Unity Feedback



Standard diagram for a servo with unity feedback

The closed-loop transfer function

K ; const.

$G(s)$; function of the complex variables

θ_i ; input

θ_o ; output

$$\varepsilon = \theta_i - \theta_o$$

$$\theta_o = KG(s)\varepsilon$$

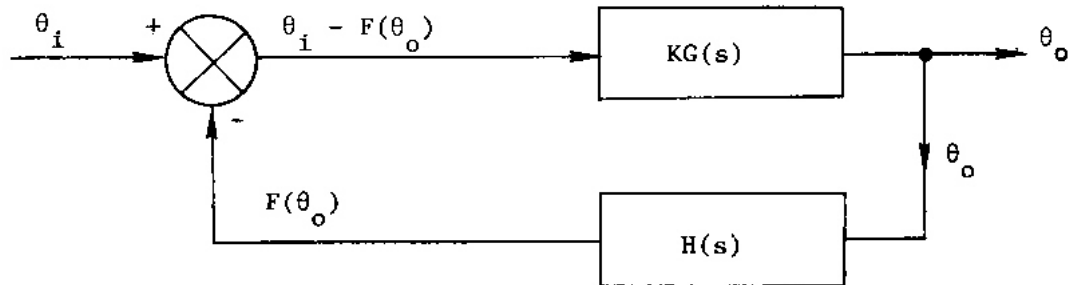
$$= KG(s)(\theta_i - \theta_o)$$

$$= KG\theta_i - KG\theta_o$$

$$(1 + KG)\theta_o = KG\theta_i$$

$$\frac{\theta_o}{\theta_i} = \frac{KG(s)}{1 + KG(s)} \quad ; \text{closed-loop transfer function}$$

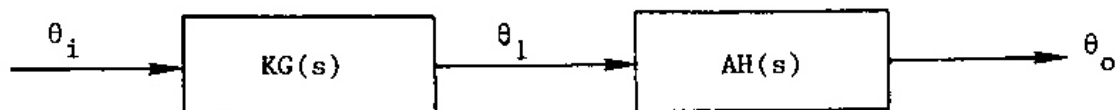
Single Loop Servo with Non-Unity Feedback



Standard diagram for a servo with non-unity feedback

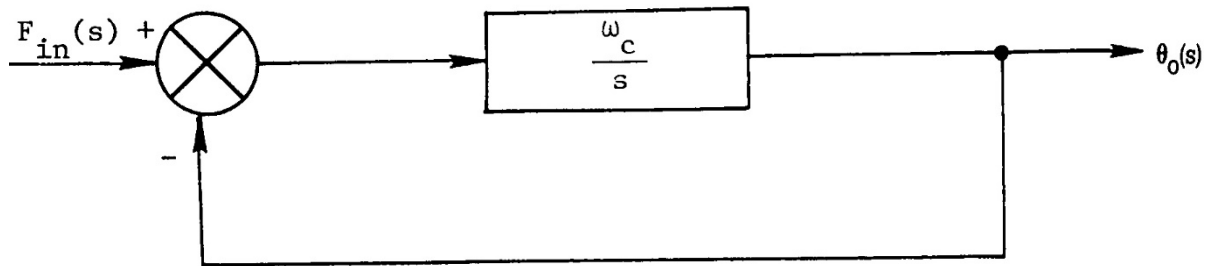
$$\begin{aligned}
 \varepsilon_0 &= \theta_i - F(\theta_o) \\
 &= \theta_i - H(s)\theta_o \\
 \theta_o &= KG(s)\varepsilon_0 \\
 &= KG(s)(\theta_i - H(s)\theta_o) \\
 (1 + KGH)\theta_o &= KG\theta_i \\
 \frac{\theta_o}{\theta_i} &= \frac{KG(s)}{1 + KG(s)H(s)}
 \end{aligned}$$

The feedback transfer function ($H(s)$) can be selected to achieve a desired transient and/or steady-state performance of the servo.



$$\begin{aligned}
 \frac{\theta_1}{\theta_i} &= KG(s) \\
 \frac{\theta_o}{\theta_1} &= AH(s) \\
 \frac{\theta_o}{\theta_i} &= KG(s)AH(s)
 \end{aligned}$$

A Simple Servomechanism



$F_{in}(s)$; a step function

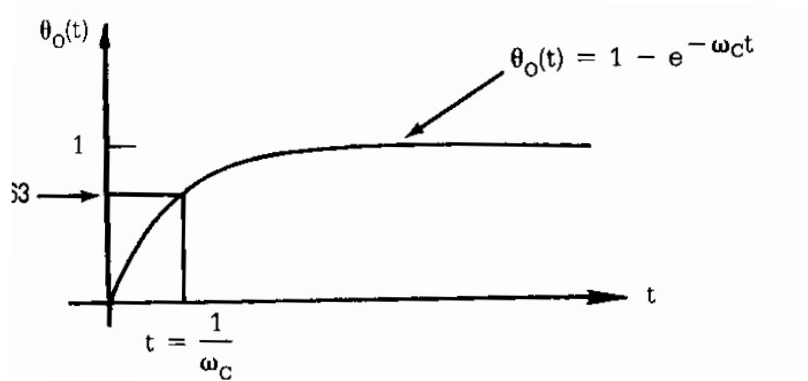
Transfer function

$$\frac{\theta_o(s)}{F_{in}(s)} = \frac{\frac{\omega_c}{s}}{1 + \frac{\omega_c}{s}} = \frac{\omega_c}{s + \omega_c}$$

$$\begin{aligned}\theta_o(s) &= \frac{\omega_c}{s + \omega_c} F_{in}(s) \\ &= \frac{\omega_c}{s + \omega_c} \frac{1}{s} \\ &= \frac{1}{s} - \frac{1}{s + \omega_c}\end{aligned}$$

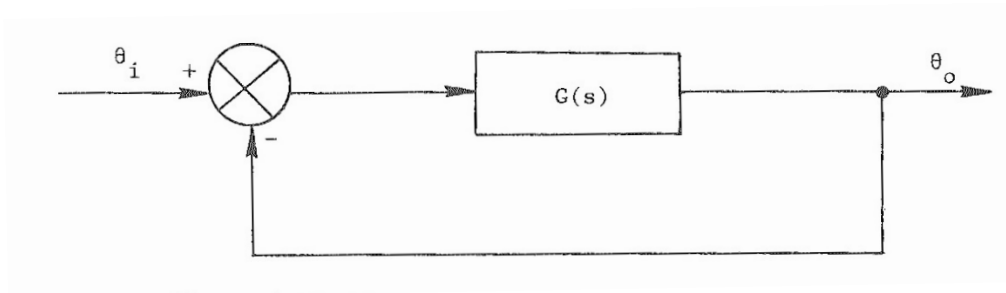
Taking the inverse Laplace transform

$$\theta_o(t) = 1 - e^{-\omega_c t}$$

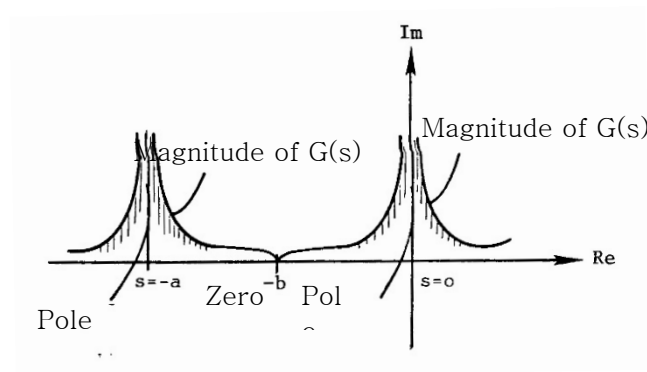


Stability

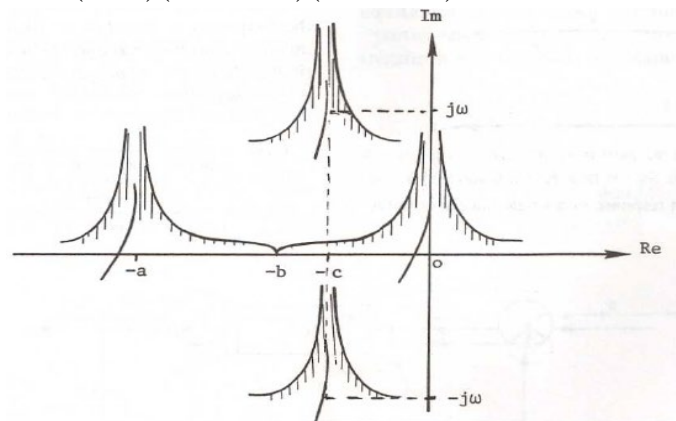
The stability of servomechanisms can be studied using frequency response plots, and the system transient response can be obtained from the roots of the characteristic equation.

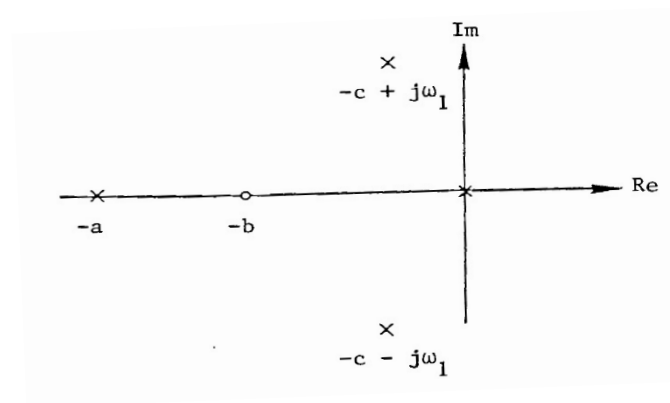


$$\text{i) } G'(s) = \frac{K(s+b)}{s(s+a)} \quad s = \sigma + j\omega$$



$$\text{ii) } G'(s) = \frac{(s+b)}{s(s+a)(s+c+j\omega)(s+c-j\omega)}$$

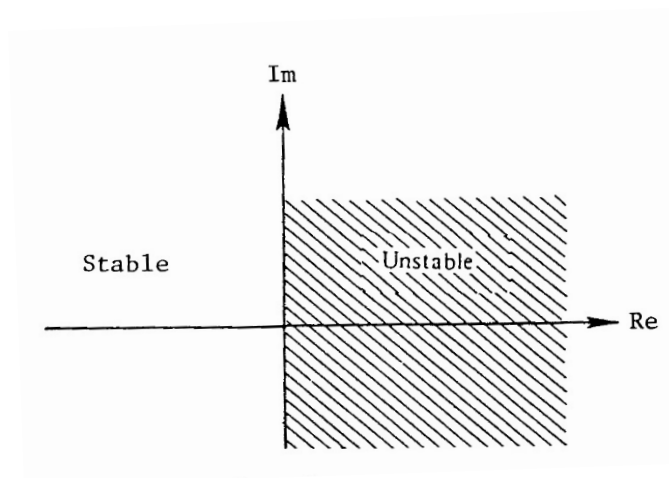




There is a direct correspondence between pole position and transient performance

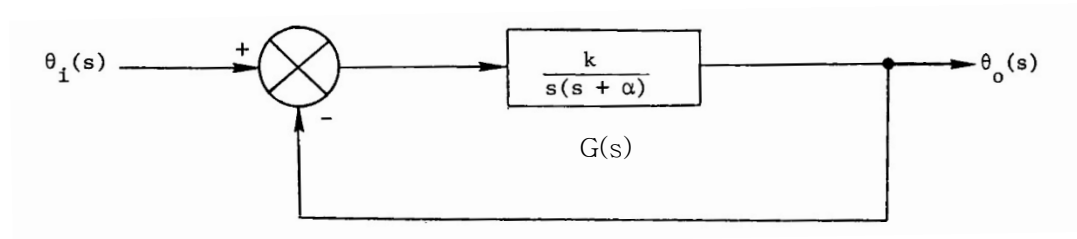
$-a$, pole $\Rightarrow e^{-at}$ response (stable)

a , pole $\Rightarrow e^{at}$ response (unstable) : it implies a continuously increasing output



Hence, the transfer function shows the stability of the given system.

s-plane Quadratic Transfer Function Response



$$G'(s) = \frac{G(s)}{1 + G(s)} = \frac{K}{s^2 + \alpha s + K}$$

where $G'(s)$ is a transfer function

$$\theta_o(s) = \theta_i(s)G'(s)$$

An input step function $\theta_i(s) = \frac{1}{s}$

$$\theta_o(s) = \frac{K}{s(s^2 + \alpha s + K)}$$

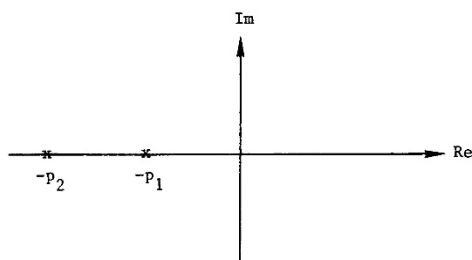
$$s^2 + \alpha s + K \Rightarrow s_{1,2} = \frac{-\alpha \pm \sqrt{\alpha^2 - 4K}}{2}$$

For $G'(s)$

Case 1: $\alpha^2 > 4K$

$$\theta_o(s) = \frac{K_1}{s} + \frac{K_2}{s + p_1} + \frac{K_3}{s + p_2}$$

$$\theta_o(t) = K_1 + K_2 e^{-p_1 t} + K_3 e^{-p_2 t}$$

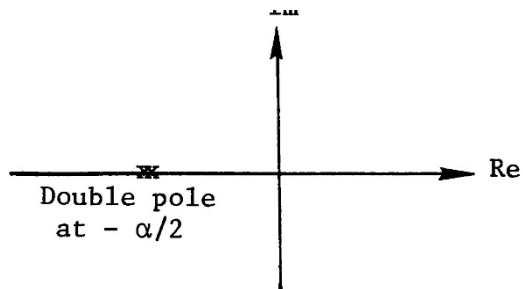


The transfer function has two real roots.

Case 2: $\alpha^2 = 4K$

$$\theta_o(s) = \frac{K_4}{s} + \frac{K_5}{\left(s + \frac{\alpha}{2}\right)^2} + \frac{K_6}{s + \frac{\alpha}{2}}$$

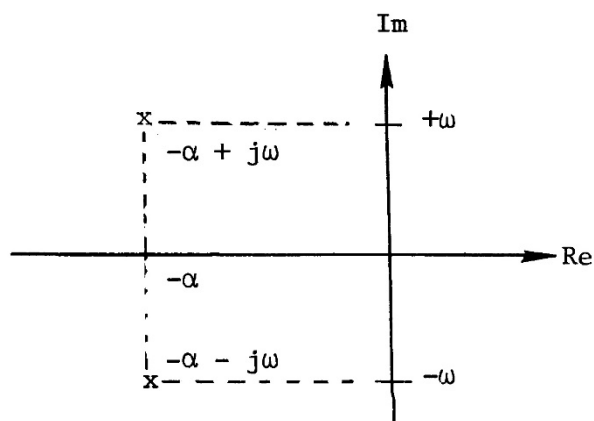
$$\theta_o(t) = K_4 + K_5 t e^{-\frac{\alpha}{2}t} + K_6 e^{-\frac{\alpha}{2}t}$$



Case 3: $\alpha^2 < 4K$

$$\theta_o(s) = \frac{K_7}{s} + \frac{K_8}{s + \alpha - j\omega} + \frac{K_9}{s + \alpha + j\omega}$$

$$\theta_o(t) = K_7 + K_{10} e^{-\alpha t} \sin(\omega t + \phi)$$



These three situations are the basic ones for a closed loop quadratic system.

Attitude Control System Example

The position (angle) and angular rates can be measured by rate and rate integrating gyros. The relationship can be expressed as: $\theta_1, \theta_2, \theta_3$ are roll, pitch yaw angle, respectively.

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} = \begin{bmatrix} 1 & \sin \theta_1 \tan \theta_2 & \cos \theta_1 \tan \theta_2 \\ 0 & \cos \theta_1 & -\sin \theta_1 \\ 0 & \sin \theta_1 / \cos \theta_2 & \cos \theta_1 / \cos \theta_2 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \quad (1)$$

Integration of eq. above results in the roll, pitch and yaw angles (errors) which must be maintained within specified limits. Combining the above kinematical relations with uncoupled dynamical equation for angular motion (Euler's eq.), a simplified single (e.g. roll) channel reaction attitude control system can be obtained.

The Laplace transform of derivative(integration) of $f(t)$

$$\text{let } F(s) = L[f(t)]$$

$$L\left[\frac{d^n f(t)}{dt^n}\right] = s^n F(s)$$

$$L\left[\int f(t) dt\right] = \frac{1}{s} F(s)$$

All the initial conditions and its derivations are equal to zero.

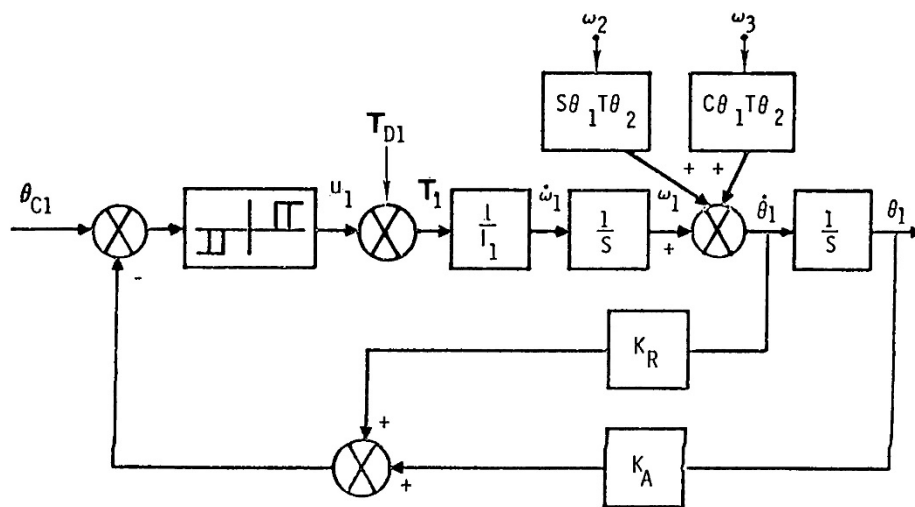
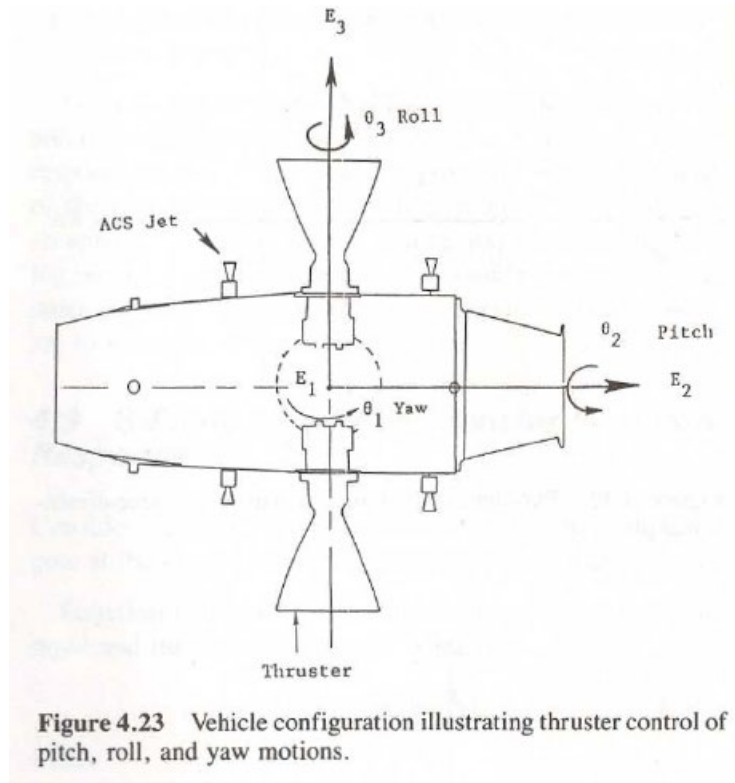


Fig 4.24 Simplified roll channel reaction attitude control system(ACS)

Both rate and position feedback signals are combined with a command signal θ_{c1} to produce a roll control torque u_1 . The latter is added to any distribution torque T_{D1} which produces the roll axis angular acceleration $\dot{\omega}_1$. Integrating $\dot{\omega}_1$ produces ω_1 which, combined with the components of ω_3 and ω_2 , yields the desired $\dot{\theta}_1$ body rate. The integral of $\dot{\theta}_1$ yields the controlled variable θ_1 , the roll angle.

To determine the position K_A and rate K_R gains and to study the performance of the system, a closed loop transfer function is required. Such a closed loop transfer function is related to θ_1 and T_{D1} . The inputs of the system are the disturbance and control torques.

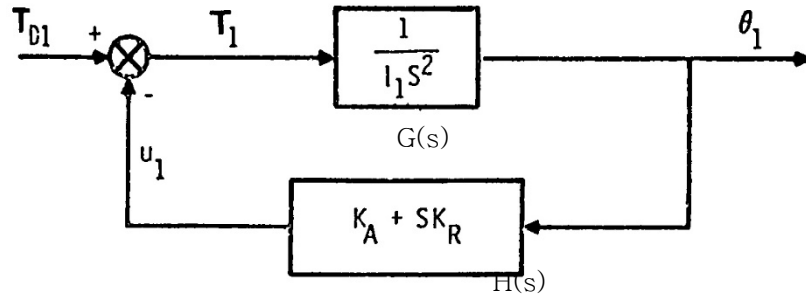


Fig 4.25 Approximation of roll channel ACS

$$\frac{\theta(s)}{T_D(s)} = \frac{G(s)}{1 + G(s)H(s)} = \frac{1}{Is^2 + K_R s + K_A} \quad (2)$$

The poles of the system are s_1 and s_2 of the characteristic equation

$$Is^2 + K_R s + K_A = 0 \quad (3)$$

$$\Rightarrow s_{1,2} = \frac{-K_R \pm \sqrt{K_R^2 - 4IK_A}}{2I}$$

The K_R , K_A values can be found if underdamped (oscillatory), critically damped, or overdamped for the dynamical system is considered.

A simple solution is the critical damping.

Then,

$$\sqrt{K_R^2 - 4IK_A} = 0 \quad \Rightarrow \quad K_A = \frac{K_R^2}{4I} \quad (4)$$

$$\Rightarrow s_{1,2} = \frac{-K_R}{2I} \equiv -\frac{1}{\tau} \quad ; \text{time const.}$$

$$K_R = \frac{2I}{\tau} \quad (5)$$

Thus,

$$\begin{aligned} \frac{\theta(s)}{T_D(s)} &= \frac{1}{Is^2 + K_R s + K_A} = \frac{1}{Is^2 + \frac{2I}{\tau}s + \frac{1}{4I} \frac{4I^2}{\tau^2}} \\ &= \frac{1}{I\left(s^2 + \frac{2}{\tau}s + \frac{1}{\tau^2}\right)} = \frac{1}{I\left(s + \frac{1}{\tau}\right)^2} = \frac{1}{I \frac{1}{\tau^2} (\tau s + 1)^2} \\ &= \frac{\tau^2}{I(\tau s + 1)^2} \end{aligned} \quad (6)$$

The performance of the system can be analyzed by examining the time domain response to a step disturbance in eq. (6).

$$T_D(s) = \frac{T_D}{s}, \quad \text{where } T_D \text{ is the magnitude of the step disturbance torque}$$

The Laplace transform of the output is

$$\theta(s) = \frac{\tau^2 T_D}{Is(\tau s + 1)^2} \quad (7)$$

$$\begin{aligned} \frac{\tau^2}{s(\tau s + 1)^2} &= \frac{1}{s\left(s + \frac{1}{\tau}\right)^2} = \frac{A}{s} + \frac{B}{\left(s + \frac{1}{\tau}\right)^2} + \frac{C}{\left(s + \frac{1}{\tau}\right)} \\ \Rightarrow A\left(s + \frac{1}{\tau}\right)^2 + Bs + Cs\left(s + \frac{1}{\tau}\right) &= 1 \\ \Rightarrow A\left(s^2 + \frac{2}{\tau}s + \frac{1}{\tau^2}\right) + Bs + Cs^2 + \frac{C}{\tau}s &= 1 \\ \Rightarrow (A + C)s^2 + \left(\frac{2}{\tau}A + B + \frac{C}{\tau}\right)s + \frac{A}{\tau^2} &= 1 \\ \Rightarrow A = \tau^2, \quad B = -\tau, \quad C = -\tau^2 \end{aligned}$$

$$\frac{\tau^2}{s(\tau s + 1)^2} = \frac{\tau^2}{s} - \frac{\tau}{\left(s + \frac{1}{\tau}\right)^2} - \frac{\tau^2}{\left(s + \frac{1}{\tau}\right)}$$

Using inverse Laplace transform, the time response of equation (7) is

$$\theta(t) = \frac{T_D}{I} \left[\tau^2 \left(1 - e^{-\frac{t}{\tau}} \right) - \tau t e^{-\frac{t}{\tau}} \right] \quad (8)$$

$$\theta(t)_{\text{maximum}} \rightarrow \theta(t)_{t=\infty} \rightarrow \frac{\tau^2 T_D}{I} \quad ; \text{steady-state value (maximum value)} \quad (9)$$

$$\theta(\infty) = \lim_{s \rightarrow 0} s \theta(s) = \lim_{s \rightarrow 0} \frac{s \tau^2 T_D}{I s (\tau s + 1)^2} = \frac{\tau^2 T_D}{I} ; \text{final value theorem}$$

If the maximum value is for controlled value $\theta_{\max}$ is specify, then the time constant τ can be obtained from eq. (9).

$$\tau = \sqrt{\frac{I \theta_{\max}}{T_D}} \quad (10)$$

From eq. (5)

$$\therefore K_R = \frac{2I}{\tau} = 2 \sqrt{\frac{I T_D}{\theta_{\max}}} \quad (11)$$

From equation(4)

$$\therefore K_A = \frac{K_R^2}{4I} = \frac{T_D}{\theta_{\max}} \quad (12)$$

The maximum control torque u which is required to limit θ to $\theta_{\max}$ can be found

$$u(s) = (K_A + s K_R) \theta(s)$$

For a step function

$$\begin{aligned}
 u(s) &= \left(\frac{I}{\tau^2} + s \frac{2I}{\tau} \right) \frac{\tau^2 T_D}{Is(\tau s + 1)^2} \\
 &= \frac{(1 + 2\tau s)T_D}{s(\tau s + 1)^2}
 \end{aligned} \tag{13}$$

By the “final value theorem”

$$\lim_{t \rightarrow \infty} u(t) = \lim_{s \rightarrow 0} su(s) = T_D$$

In the limit, the control torque must be equal to the disturbance torque T_D

From inverse Laplace Transform we have

$$u(t) = T_D \left[\left(1 - e^{-\frac{t}{\tau}} \right) - \frac{t}{\tau} e^{-\frac{t}{\tau}} \right] \tag{14}$$

Note: the control torque implied to the system is $-u(t)$ (see Fig. 4.25).